

Bird Responses to a Managed Forested Landscape

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Natural History of the Red Owl (*Tyto soumagnei*) in Dry Deciduous Tropical Forest in Madagascar

Scott G. Cardiff^{1,3,4} and Steven M. Goodman²

ABSTRACT.—Recent observations of the Red Owl (*Tyto soumagnei*) in Madagascar demonstrated that it inhabits dry deciduous forest, and roosts on rock ledges and in cave entrances in the extreme north of the island. We observed a Red Owl at a sinkhole site in the Réserve Spéciale d'Ankarana, found evidence of its use of an additional cave, and collected its pellets in three separate dry seasons between 2000 and 2003. Tsingy tufted-tailed rats (*Eliurus antsingy*) constituted almost 50% of the total prey mass of Red Owls at Ankarana. Their diet at Ankarana differed from that of Red Owls from Masoala in the humid northeast of Madagascar, as the Ankarana pellets contained insects, frogs, and numerous geckos. Red Owls appear to consume more native than introduced rodents and do not appear to prey upon birds or bats like other large owls on the island. Forest degradation could reduce densities of tufted-tailed rats and could be a conservation threat to this owl. Received 30 May 2006. Accepted 7 February 2008.

The Red Owl (*Tyto soumagnei*) is rare, endemic to Madagascar and was not observed by biologists for several decades (Thorstrom and Rene de Roland 2003). It is considered "Endangered" (IUCN 2006) and was thought to occur only in humid forests in the east (Grandidier 1878, Halleux and Goodman 1994, Goodman et al. 1996, Thorstrom et al. 1997, ZICOMA 1999, Irwin and Samonds 2002). Limited diet studies suggest the owl feeds on frogs, geckos, afrosoricidans (*Microgale* spp., *Oryzorictes hova*), rodents (*Eliurus*

spp., *Rattus rattus*), and the eastern rufous mouse lemur (*Microcebus rufus*) (Lavauden 1932, Halleux and Goodman 1994, Goodman and Thorstrom 1998). These different prey animals are native to Madagascar with the exception of the roof rat (*Rattus rattus*). Until recently, observers have noted owls roosting only in trees in intact and degraded humid forest (Thorstrom and Rene de Roland 1997, Thorstrom et al. 1997, Irwin and Samonds 2002).

Cardiff and Befourouack (2003) and Van Esbroeck (2006) extended the known range of the owl to Ankarana in the north of the island. Van Esbroeck (2006) observed Red Owls at Ankarana in 1999; that observation represents the first record of the occurrence of this owl in dry deciduous forest and the first record of it roosting in a cave entrance or rocky crevice. Our observation of a Red Owl at Ankarana in 2000 (Cardiff and Befourouack 2003) prompted us to obtain further information on the distribution, diet, and roosting habits of this poorly known species and its congener, the Barn Owl (*T. alba*) at Ankarana.

METHODS

The Réserve Spéciale d'Ankarana, ~60 km south of Antsiranana in northern Madagascar (Fig. 1) (12° 49' to 13° 01' S and 49° 00' to 49° 16' E), encompasses a massif of Jurassic limestone with dry deciduous forest partially surrounded by and intruded upon by semi-deciduous forest (Cardiff and Befourouack 2003). Precipitation near the reserve averages 180–200 cm per year (Rossi 1976, Hawkins et al. 1990). The local limestone karst features many caves and several river sinkholes (Radofilao 1977). These include, on the east side of the reserve, a large collapse sinkhole (collapse doline), where Van Esbroeck (2006) observed a Red Owl in 1999 at the Perte des Rivières, which is also known as the Perte de la Besaboba Number 1 (Radofilao 1977; J.

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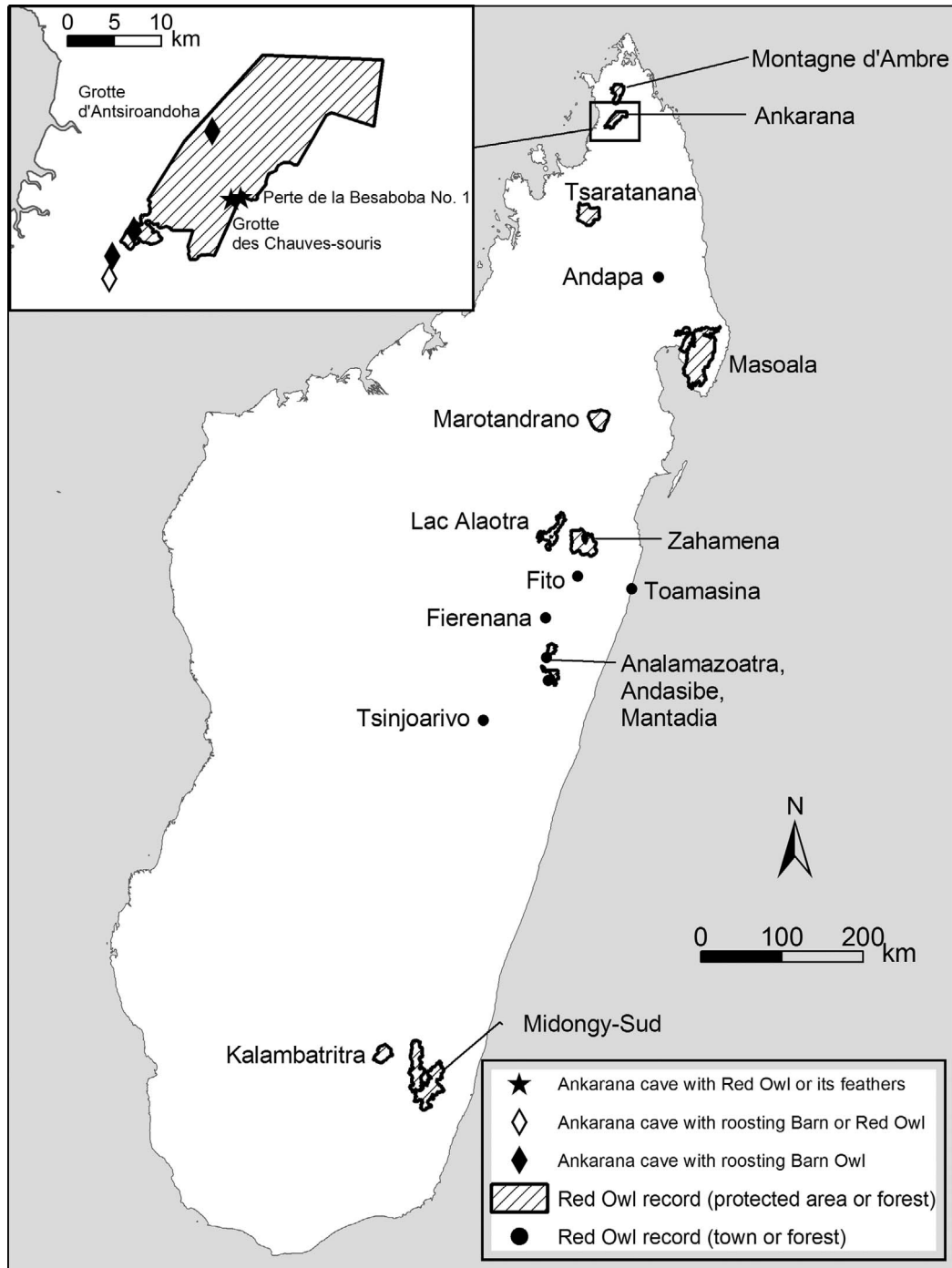


FIG. 1. Recorded distribution of *Tyto soumagnei* on Madagascar and in Ankarana, and distribution of *Tyto alba* at Ankarana.

Radofilao, unpubl. data) and “Perte des trois rivières” (Van Esbroeck 2006). Savannah occurs within ~200 m and semi-deciduous forest on basalt occurs within 500 m of this sink-hole site.

We recorded the presence or absence of all owl species seen or heard at 28 caves from 2003 to 2006 in and around Ankarana during cave biological inventory and monitoring work (Cardiff 2006). We made additional observations at the Perte des Rivières site in the 2000, 2001, and 2003 dry seasons between June and November. We noted the presence of pellets and checked potential roost positions for owls on several occasions to understand roost characteristics and periodicity of roost occupation in 2000 and 2001, and also collected pellets in 2003. We collected >45 pellets and pellet fragments at this site over the three dry seasons. We assumed that all pellets found at the site were from Red Owls based on the concurrent observation of an individual present or feathers attributable to the species on two of three collection dates in 2000 and on two of five dates in 2001. We also collected broken pellets at the entrances of the Grotte d’Antsironandoha on the western side of Ankarana. We attributed pellets collected at Grotte d’Antsironandoha to Barn Owls because we observed a Barn Owl at this location. We did not conduct formal inventories for owls in forests away from cave entrances.

We measured the dimensions of several intact pellets and identified prey items in all collected material. We washed the pellet contents through a sieve after soaking pellets in soapy water and identified recovered hard items to species or species group using collections material at the Field Museum of Natural History in Chicago. We counted the minimum number of individuals (MNI) by the number of unique bilateral elements for each species or species group (Goodman et al. 1991), and calculated relative contribution to total number of individuals and relative contribution to total prey mass based on reference prey masses (Marti 1987).

RESULTS

Distribution.—We observed *Tyto* spp. at five of 28 caves. Three of those observations were of a Barn Owl and one was not identified to species. We only observed Red Owls at the Perte des Rivières, but we also observed and

photographed a feather of a Red Owl in 2006 at the entrance to the Grotte des Chauves-souris (“Andavampanihy”), a large cave 1 km west of the Perte des Rivières.

Roosting Habits.—On 31 July 2000 at 0800 hrs, SGC observed one Red Owl on a ledge in a crack in the limestone below the overhang of the edge of the Perte des Rivières sinkhole. SGC again observed one individual at the same location on 6 August 2000 and again on 3 September 2000. SGC subsequently observed an owl at the site on 1 July 2001 and 5 August 2001 but on different rock ledges. Feathers of Red Owls were also present with collected pellets on 12 November 2000. We did not detect the owl at this site during 16 additional visits in 2000–2001.

The Red Owl was roosting in the general area of the opening to the lower passage in the sinkhole in all observations. The owl was in a fissure about 0.5×0.5 m in the ceiling (~8 m from the floor) in front of the entrance to the horizontal passage under the overhang when observed in 2000. Guides had reportedly previously seen two Red Owls together at this location. The owl was roosting on the first occasion in 2001 on a ledge about 3 m above the cave floor and 5 m into the horizontal passage. On another occasion, it was roosting in a small opening on the side of the sinkhole at the entrance to the horizontal passage, ~1.5 m above the floor.

Food Habits.—Four intact Red Owl pellets measured 35×24 , 40×24 , 41×32 , and 58×29 mm (length $\times$ width). Pellets contained a minimum number of individuals (MNI) of 172 insects, frogs, geckos, and mammals (Table 1). Mammals constituted the greatest proportion of the prey items by individual and by mass for all years with geckos being important in 2000. Frogs, a primate, and insects were present in the pellets in some years but not in others (Table 1). The Madagascan pygmy shrew (*Suncus madagascariensis*) contributed most to the MNI in 2001 and the roof rat contributed up to 27% of the total prey biomass in a given year. Tsingy tufted-tailed rats (*Eliurus antsingy*) contributed the most of any single prey item to prey mass in all years and to prey total MNI in 2000 and 2003 (Table 1).

Barn Owl Pellets from Grotte d’Antsironandoha contained frogs (MNI = 4), birds (*Turnix nigricollis*, MNI = 1; *Foudia*

TABLE 1. Relative importance of tsingy tufted-tailed rats as prey items of one or more Red Owls at Ankarana, Madagascar based on regurgitated pellet samples from three different years. Possible frog and gecko prey species identification is based on records of herpetological collections (A. Raselimanana and M. Razafim-pahanana, unpubl. data; C. Raxworthy, unpubl. data). PN = number of entire pellets (* additional fragments also examined). MNI = Minimum Number of Individuals.

Prey item	Mass (g)	Collection year (and months), collection totals, and percent of totals for prey items							
		2000 (Sep–Nov)		2001 (Jun–Sep)		2003 (Jun–Jul, Nov)		All years combined	
		PN = 8*	MNI = 30	PN = 25	MNI = 90	PN = 12*	MNI = 52	PN = 45*	MNI = 172
		MNI (%)	Mass (%)	MNI (%)	Mass (%)	MNI (%)	Mass (%)	MNI (%)	Mass (%)
Insecta				8.9	0.8	5.8	0.8	6.4	0.6
Coleoptera	~4 ^h			2.2	0.3	3.8	0.4	2.3	0.2
Dictyoptera: Blaberidae ^a	7.5 ⁱ					1.9	0.4	0.6	0.1
Orthoptera	~2.5 ^h			6.7	0.5			3.5	0.2
Amphibia: Anura						17.3	2.8	5.2	0.8
Large ^b	~9.5 ^j					5.8	1.5	1.7	0.4
Medium ^c	~4.5 ^k					7.7	1.0	2.3	0.3
Small ^d	~3.0 ^j					3.8	0.3	1.2	0.1
Squamata: Gekkonidae			19.8	24.4	24.3	28.8	26.7	27.9	23.8
Large ^e	~50 ^l	10.0	8.4	7.8	11.7	7.7	10.6	8.1	10.5
Medium ^f	~29 ^m	20.0	9.7	12.2	10.6	19.2	15.3	15.7	11.7
Small ^g	~15 ^l	6.7	1.7	4.4	2.0	1.9	0.8	4.1	1.6
Mammalia		63.3	80.2	66.7	74.9	48.1	69.7	60.5	74.9
Soricomorpha		10.0	0.4	36.7	2.5	17.3	1.5	26.2	1.7
<i>Suncus madagascariensis</i>	2.3 ^k	10.0	0.4	36.7	2.5	15.4	1.0	25.6	1.5
Afrosoricida						1.9	0.6	0.6	0.2
<i>Microgale breviceaudata</i>	10.7 ^l					1.9	0.6	0.6	0.2
Rodentia		53.3	79.9	30.0	72.4	25.0	58.5	32.6	70.5
<i>Eliurus antsingy</i>	87.5 ^l	46.7	68.4	16.7	43.7	17.3	41.7	22.1	49.8
<i>Mus musculus</i>	9.7 ^k			4.4	1.3	1.9	0.5	2.9	0.7
<i>Rattus rattus</i>	102.7 ^k	6.7	11.5	8.9	27.4	5.8	16.3	7.6	20.0
Primata						5.8	9.7	1.7	2.7
<i>Microcebus tavaratra</i>	61.1 ⁿ					5.8	9.7	1.7	2.7

^a *Gromphadorhina* sp.

^b cf. *Laliostoma* sp./*Mantidactylus* spp./*Aglyptodactylus* sp.

^c cf. *Boophis* sp./*Mantidactylus* spp./*Ptychadena madagascariensis*.

^d *Mantella* spp./*Anodontohyla* sp./*Cophyla* sp./*Stumpffia* sp.

^e cf. *Homopholis boivini*/*Uroplatus fimbriatus*/*U. henkeli*.

^f cf. *Uroplatus sikorae*/*Paroedura homalorhina*/*P. stumpffii*.

^g cf. *U. ebenau*/*P. karstophila*/*P. oviceps*/*Geckolepis* spp./*Hemidactylus* spp.

^h Estimated.

ⁱ Clarke and Moore (1995).

^j Estimated (Goodman et al. 1993b).

^k Goodman et al. (1993b).

^l SMG, unpubl. data.

^m Goodman et al. (1991).

ⁿ Rasoloarison et al. (2000).

madagascariensis, MNI = 1; *Merops superciliosus*, MNI = 1), afrosoricidans and soricomorphans (*Setifer setosus*, MNI = 1; *Suncus madagascariensis*, MNI = 1), rodents (*Mus musculus*, MNI = 8; *Rattus rattus*, MNI = 28), and a northern rufous mouse lemur (*Microcebus tavaratra*, MNI = 1).

DISCUSSION

This work provides new information on the habits of Red Owls in a forest biome in which it was only recently found to occur. Our find-

ings add to knowledge of the distribution, roosting habits, and diet of this rare owl, which has implications for their conservation at Ankarana.

The Ankarana record is the only published account of Red Owls using a cave (Van Esbroek 2006). However, the occurrence of the owl at a different nearby cave suggests it is not an unusual roost site. The continued presence of at least one individual at the site in the dry season of several successive years (1999, 2000, 2001, 2003) suggests the use of

the first site is not incidental. An additional unpublished record of a Red Owl using a cave entrance for roosting elsewhere on the island suggests these owls use caves for roosting more often than previously thought. In 2003, A. Raselimanana and SMG found a single Red Owl roosting in a shallow cave in the Mt. Papango portion of the Parc National de Midongy-Sud (Fig. 1) in an extensive area of humid forest in the southeastern portion of the island. This record is important because it extends the known range of the species to the south (Irwin and Samonds 2002) and shows that it uses caves for roosting in the eastern portion of its range. Use of cave entrances suggests a large number of potential roost sites could be available for Red Owls at Ankarana, a zone of extensive karst, and that Red Owls could occur in other areas with caves or rock crevices. However, Barn Owls appear to use cave entrances more commonly than Red Owls at Ankarana based on our observations.

Our data also expands knowledge of food habits of Red Owls. Early observations suggested this species fed extensively on frogs (Lavauden 1932) and a captive individual fed on frogs (Halleux and Goodman 1994). No anurans occurred in pellets from the Masoala Peninsula (Goodman and Thorstrom 1998). Frogs were present in the Ankarana pellets even though frogs are more abundant on the Masoala Peninsula than at the notably drier Ankarana. Goodman and Thorstrom (1998) found no insects, no *Suncus* spp., and few geckos in pellets from the Masoala Peninsula. We found some insects and many Madagascan pygmy shrews and geckos in pellets from Ankarana. Mean prey weight was lower at Ankarana than on the Masoala Peninsula due to the presence of insects, frogs, and smaller-bodied afrosoricidans in the pellets. The lack of bats or birds, and the importance of tufted-tailed rats and native mammals in the diet of Red Owls remained consistent between Ankarana and Masoala, and appear to distinguish the dietary regime of Red Owls from that of Barn Owls, including at Ankarana, and the Madagascar Owl (*Asio madagascariensis*) on the island (Goodman and Langrand 1993; Goodman and Thorstrom 1998; Goodman et al. 1993a,b; Goodman and Griffiths 2006; Rasoma and Goodman 2007; this study). The absence of birds or bats in the diet of Red Owls

is notable given the presence of at least one species of bat in the cave where the owl roosted at Ankarana (Cardiff 2006), and the presence of birds and bats (*Hipposideros comersoni*, *Coleura afra*) (Goodman et al. 2008) in the diet of Barn Owls at Ankarana. Unlike diet, pellet dimensions do not seem to distinguish Red Owls at Ankarana ($35\text{--}58 \times 24\text{--}32$ mm, $n = 4$) from Barn Owls at other sites on Madagascar ($37\text{--}62 \times 21\text{--}33$ mm, $n = 23$) (Goodman et al. 1993b) and identification of pellet species origin by size alone is not appropriate.

CONSERVATION IMPLICATIONS

Habitat degradation may threaten Red Owls at Ankarana. This species roosts and feeds in degraded areas on the Masoala Peninsula; the roost at Ankarana is a rocky area surrounded by forest edge not far from degraded forest and savannah. The owls at both sites appear to feed primarily on tufted-tailed rats. Tufted-tailed rats often climb vegetation and trees, do not seem to occur in savannahs, and rarely occur in areas with severely degraded vegetation (Carleton 2003). Ongoing degradation of forests at Ankarana may reduce availability of tsingy tufted-tailed rats as a food source for Red Owls. Forest degradation could also reduce availability of large trees with cavities that remain the only confirmed nest site for this owl (Thorstrom and Rene de Roland 1997), although it is likely that Red Owls nest in rock cavities since it is now known to roost in such cavities.

Additional conservation concerns exist for the Perte des Rivières site at Ankarana. Sediments and debris borne by floodwaters during the wet season, most likely resulting from cyclones and upstream deforestation, have previously blocked most of the length of horizontal cave passage at the roost site (S. G. Cardiff, unpubl. data). The potential effect of blocking of the cave by sediment and debris on the suitability of the site for roosting by Red Owls is unknown. The site is also accessible to tourists and, although descending into the sinkhole is dangerous without proper equipment and expertise, some tourist visits to the bottom occur. Repeated disturbances by tourists or others could possibly cause Red Owls to abandon use of the site. We encourage other studies to examine potential threats

to conservation of Red Owls and interactions between Red Owls and Barn Owls.

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LITERATURE CITED

- CARDIFF, S. G. 2006. Bat cave selection and conservation in Ankarana, northern Madagascar. Thesis. Columbia University, New York, USA.
- CARDIFF, S. G. AND J. BEFOUROUACK. 2003. The Réserve Spéciale d'Ankarana. Pages 1501–1507 in *The natural history of Madagascar* (S. M. Goodman and J. P. Benstead, Editors). University of Chicago Press, Chicago, Illinois, USA.
- CARLETON, M. D. 2003. *Eliurus*, tufted-tailed rats. Pages 1373–1380 in *The natural history of Madagascar* (S. M. Goodman and J. P. Benstead, Editors). University of Chicago Press, Chicago, Illinois, USA.
- CLARKE, D. C. AND A. J. MOORE. 1995. Variation and repeatability of male agonistic hiss characteristics and their relationship to social rank in *Gromphadorhina portentosa*. *Animal Behaviour* 50:719–729.
- GOODMAN, S. M. AND O. GRIFFITHS. 2006. A case of exceptionally high predation levels of *Rousettus madagascariensis* by *Tyto alba* (Aves: Tytonidae) in western Madagascar. *Acta Chiropterologica* 8: 553–556.
- GOODMAN, S. M. AND O. LANGRAND. 1993. Food habits of the Barn Owl *Tyto alba* and the Madagascar Long-eared Owl *Asio madagascariensis* on Madagascar: adaptation to a changing environment. *Musée Royal de l'Afrique Centrale Tervuren, Belgique, Annales Sciences Zoologiques* 268:147–153.
- GOODMAN, S. M. AND R. THORSTROM. 1998. The diet of the Madagascar Red Owl (*Tyto soumagnei*) on the Masoala Peninsula, Madagascar. *Wilson Bulletin* 110:417–421.
- GOODMAN, S. M., S. G. CARDIFF, AND F. H. RATRIMOMANARIVO. 2008. First record of *Coleura* (Emballonuridae) on Madagascar and identification and diagnosis of members of the genus. *Systematics and Biodiversity* 6:283–292.
- GOODMAN, S. M., G. K. CREIGHTON, AND C. RAXWORTHY. 1991. The food habits of the Madagascar Long-eared Owl *Asio madagascariensis* in south-eastern Madagascar. *Bonner zoologische Beiträge* 42:21–26.
- GOODMAN, S. M., O. LANGRAND, AND C. J. RAXWORTHY. 1993a. Food habits of the Madagascar Long-eared Owl *Asio madagascariensis* in two habitats in southern Madagascar. *Ostrich* 64:79–85.
- GOODMAN, S. M., O. LANGRAND, AND C. J. RAXWORTHY. 1993b. The food habits of the Barn Owl *Tyto alba* at three sites on Madagascar. *Ostrich* 64:160–171.
- GOODMAN, S. M., A. ANDRIANARIMISA, L. E. OLSON, AND V. SOARIMALALA. 1996. Patterns of elevational distribution of birds and small mammals in the humid forests of Montagne d'Ambre, Madagascar. *Ecotropica* 2:87–98.
- GRANDIDIER, A. 1878. Note sur un nouveau Strigidé de Madagascar. *Bulletin de la Société philomathique de Paris* 7(2):65–66.
- HALLEUX, D. AND S. M. GOODMAN. 1994. The rediscovery of the Madagascar Red Owl *Tyto soumagnei* (Grandidier 1878) in north-eastern Madagascar. *Bird Conservation International* 4:305–311.
- HAWKINS, A. F. A., P. CHAPMAN, J. U. GANZHORN, Q. C. M. BLOXAM, S. C. BARLOW, AND S. J. TONGE. 1990. Vertebrate conservation in Ankarana Special Reserve, northern Madagascar. *Biological Conservation* 54:83–110.
- IRWIN, M. T. AND K. E. SAMONDS. 2002. Range extension of the Madagascar Red Owl *Tyto soumagnei* in Madagascar: the case of a rare, widespread species? *Ibis* 144:680–683.
- INTERNATIONAL UNION FOR THE CONSERVATION OF NATURE (IUCN). 2006. IUCN Red List of threatened species. International Union for Conservation of Nature and Natural Resources Species Survival Commission, Cambridge, United Kingdom. <http://www.iucnredlist.org> (accessed 17 May 2006).
- LAVAUDEN, M. L. 1932. Etude d'une petite collection d'oiseaux de Madagascar. *Bulletin du Muséum national d'Histoire naturelle* (2) 4:629–640.
- MARTI, C. D. 1987. Raptor food habits studies. Pages 67–80 in *Raptor management techniques manual* (B. A. Giron Pendleton, B. A. Millsap, K. W. Cline, and D. M. Bird, Editors). National Wildlife Federation, Washington, D.C., USA.
- RADOFILAO, J. 1977. Bilan des explorations spéléologiques dans l'Ankarana. *Annales de l'Université de Madagascar, série Sciences de la Nature et Mathématiques* 14:195–204.
- RASOLOARISON, R., S. M. GOODMAN, AND J. U. GAN-

- ZHORN. 2000. Taxonomic revision of mouse lemurs (*Microcebus*) in the western portions of Madagascar. *International Journal of Primatology* 21:963–1019.
- RASOMA, J. AND S. M. GOODMAN. 2007. Food habits of the Barn Owl (*Tyto alba*) in spiny bush habitat of arid southwestern Madagascar. *Journal of Arid Environments* 69:537–543.
- ROSSI, G. 1976. Karst et dissolution des calcaires en milieu tropical. *Zeitschrift für Geomorphologie, Supplementbände* 26:124–152.
- THORSTROM, R. AND L.-A. RENE DE ROLAND. 1997. First nest record and nesting behaviour of the Madagascar Red Owl *Tyto soumagnei*. *Ostrich* 68: 42–43.
- THORSTROM, R. AND L.-A. RENE DE ROLAND. 2003. *Tyto soumagnei*, Madagascar Red Owl. Pages 1110–1113 in *The natural history of Madagascar* (S. M. Goodman and J. P. Benstead, Editors). University of Chicago Press, Chicago, Illinois, USA.
- THORSTROM, R., J. HART, AND R. T. WATSON. 1997. New record, ranging behaviour, vocalization and food of the Madagascar Red Owl *Tyto soumagnei*. *Ibis* 139:477–481.
- VAN ESBROECK, J. 2006. Madagascar Red Owl *Tyto soumagnei* in Ankarana Special Reserve, Madagascar. *Bulletin of the African Bird Club* 13:205–206.
- ZICOMA. 1999. Les Zones d'Importance pour la Conservation des Oiseaux à Madagascar. *Projet ZICOMA*, Antananarivo, Madagascar.

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Bird Responses to a Managed Forested Landscape

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ABSTRACT.—I examined bird population responses to a managed forested landscape resulting from management for Ruffed Grouse (*Bonasa umbellus*) habitat in central Pennsylvania during three consecutive springs, 2005–2007. The number of bird species increased from 2001–2002 ($n = 40$) to 2005–2007 ($n = 46$). Abundance of all species combined declined ($0.10 \geq P \geq 0.05$), perhaps because the area was more heterogenous in 2001–2002 than in 2005–2007. Red-eyed Vireo (*Vireo olivaceus*) was the most common species in both 2001–2002 and 2005–2007. Six of the 20 common species were detected only in the treated sector in both periods; none was specific to the reference sector. Despite increased forest maturation, no populations of early successional bird species declined ($P \leq 0.05$) between periods, but populations of three other species did. Management of the Barrens Grouse Habitat Management Area for Ruffed Grouse habitat did not have a profound effect on bird populations from 2001–2002 to 2005–2007 subsequent to the last cutting cycle. *Received 16 January 2008. Accepted 1 May 2008.*

Long-term studies of the responses of birds to managed landscapes are crucial to understanding regional population trends (Gullion 1990; Yahner 2000, 2003). Thus, an important question is not necessarily can habitat be man-

aged for early successional bird species (Yahner 2003), but how soon after management must habitat be manipulated further to ensure adequate early successional habitat is available due to loss via plant succession.

The Barrens Grouse Habitat Management Area (GHMA) was created in 1976 for Ruffed Grouse (*Bonasa umbellus*) habitat via even-aged management (Yahner 1993). Habitat management consisted of the establishment of 4-ha “activity centers” for adult grouse (Gullion 1977), which resulted in a series of 1-ha contiguous plots of different ages in mixed-oak (*Quercus* spp.) and aspen (*Populus* spp.) cover types (Yahner 1993). Grouse habitat management provided valuable breeding habitat for bird species dependent on young forests. The objective of the present study was to compare community structure and composition of breeding birds in the present (2005–2007) 5 years after a fourth cutting cycle (in winters 1999–2000) at the Barrens GHMA to that just after the fourth cycle in 2001–2002 (Yahner 2003).

METHODS

Field work was conducted during three consecutive springs 2005, 2006, and 2007 (late May–mid-Jun) at the Barrens GHMA, which consists of 1,120 ha on State Game Lands

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176, Centre County, in the Valley and Ridge Province of Pennsylvania (Yahner 1993) (40° 47' N, 78° 58' W). The Barrens GHMA is divided into a treated (managed) sector and a reference (unmanaged) sector of approximately equal area.

The treated sector was 136 contiguous blocks (60 in aspen and 76 in mixed-oak cover type); each block was 4 ha and was subdivided into 1-ha plots (A–D, each ~100 × 100 m) (Yahner 2003). The 60 blocks in aspen cover type were cut in four winter cutting cycles: 1976–77, plot A; 1980–81, plot B; 1986–87, plot C; and 1999–2000, plot D, for a 100% cut. Only plots A–C were cut in the 76 blocks of the mixed-oak type: 1976–77, plot A; 1980–81, plot B; and 1999–2000, plot C, for a 75% cut. Fifteen–20 residual overstory trees (woody stem >7.5 cm dbh, >1.5-m tall) were retained per plot in plots C and D of the aspen type, and in plots B and C of the mixed-oak type.

Overstory trees in the uncut plots (D in mixed oak) of the treated sector and in the reference (unmanaged) sector were ~90–100 years old and consisted mainly of red maple (*Acer rubrum*), white oak (*Quercus alba*), chestnut oak (*Q. montana*), northern red oak (*Q. rubra*), black cherry (*Prunus serotina*), quaking aspen (*Populus tremuloides*), big-tooth aspen (*P. grandidentata*), and pitch pine (*Pinus rigida*) (Yahner 1993, 2003). Major understory trees (woody stem 1–7.5 cm dbh, >1.5-m tall) were red maple, oak, black cherry, and aspen.

I used the same 90 representative plots as in previous studies (Yahner 1993, 1997, 2003). Briefly, 10 plots (A–D) in both cover types of the treated sector ($n = 80$ plots) and 10 plots in the control sector were selected. Plots were separated by 200 m and plot centers were >50 m from disturbances (e.g., logging roads) to minimize edge effects (Strelke and Dickson 1980, Yahner 1987). Area sampled was 25.4 ha or 2.3% of the total study area.

I visited each plot in late May to early June each year (2005–2007) in a randomized order (Yahner 2003). All birds seen or heard within 30 m of each plot center during a 5-min period after a 1-min equilibrium period were counted. Point counts were made between sunrise and 0900 hrs EST. A 30-m radius point count

was used to minimize edge effects at interfaces of plots of different age post-cutting in the treated sector (Repenning and Labisky 1985). Birds flying over a plot were not counted, and movements of individual birds were monitored carefully to reduce the likelihood of counting the same bird more than once in the same morning.

I calculated species richness and total abundance of each species in 2005–2007 combined in each of the three sectors (aspen = 100% clearcut; mixed-oak = 75% clearcut, and reference = 0% clearcut). I compared observed versus expected abundances of all bird species combined and of common species (≥ 10 contacts in either 2001–2002 or 2005–2007), using Chi-square goodness-of-fit tests (Sokal and Rohlf 1995). I calculated expected number of contacts in 2001–2002 or in 2005–2007 as the total number of contacts observed per period divided by 2 or 3 years, respectively (after Yahner 2003).

RESULTS

The number of species increased from 2001–2002 ($n = 40$) to 2005–2007 ($n = 46$) (Table 1). Abundance of all species combined declined ($0.10 \geq P \geq 0.05$) between periods. The most common species in both periods was Red-eyed Vireo (*Vireo olivaceus*). Six of the 20 common species were found only in the treated sector in both periods; none was specific to the reference sector.

Species present on the area in 2001–2002 but not in 2005–2007 were Cooper's Hawk (*Accipiter cooperii*), Ruby-throated Hummingbird (*Archilochus colubris*), Olive-sided Flycatcher (*Contopus cooperi*), Willow Flycatcher (*Empidonax traillii*), and Yellow-breasted Chat (*Icteria virens*). Species absent on the area in 2001–2002 but observed in 2005–2007 were Red-tailed Hawk (*Buteo jamaicensis*), Baltimore Oriole (*Icterus galbula*), Worm-eating Warbler (*Helmitheros vermivorus*), Black-throated Green Warbler (*Dendroica virens*), Blue-winged Warbler (*Vermivora pinus*), Mourning Warbler (*Oporornis philadelphia*), Hooded Warbler (*Wilsonia citrina*), Least Flycatcher (*Empidonax minimus*), Brown Thrasher (*Toxostoma rufum*), Wild Turkey (*Meleagris gallopavo*), and Vee-ry (*Catharus fuscescens*).

Population densities of Gray Catbird (*Du-*

TABLE 1. Population abundance (contacts/100 ha/year) of common species (≥ 10 contacts during springs (late May–mid-Jun) in either 2001–2002 or 2005–2007) at the Barrens Grouse Habitat Management Area (actual number of contacts in parentheses). Contacts of uncommon species are included in values of species richness and total population abundances of all species combined.

	2001–2002	2005–2007
Blue Jay, <i>Cyanocitta cristata</i> ^a	26 (13)	10 (8)
Tufted Titmouse, <i>Baeolophus bicolor</i> ^b	10 (5)	22 (17)
Black-capped Chickadee, <i>Poecile articipillus</i>	20 (10)	16 (12)
Blue-gray Gnatcatcher, <i>Poliophtila caerulea</i>	18 (9)	14 (11)
Wood Thrush, <i>Hylocichla mustelina</i> ^c	20 (10)	9 (7)
Gray Catbird, <i>Dumetella carolinensis</i> ^{ac}	35 (18)	83 (63)
Eastern Wood-Pewee, <i>Contopus virens</i>	14 (7)	13 (10)
Red-eyed Vireo, <i>Vireo olivaceus</i>	162 (82)	161 (123)
Chestnut-sided Warbler, <i>Dendroica pensylvanica</i> ^{ac}	32 (16)	92 (70)
Ovenbird, <i>Seiurus aurocapilla</i> ^a	122 (62)	69 (53)
Common Yellowthroat, <i>Geothlypis trichas</i> ^c	57 (29)	39 (30)
Hooded Warbler, <i>Wilsonia citrina</i> ^a	0 (0)	13 (10)
American Redstart, <i>Setophaga ruticilla</i> ^a	65 (33)	34 (26)
Rose-breasted Grosbeak, <i>Pheucticus ludovicianus</i>	33 (17)	39 (30)
Indigo Bunting, <i>Passerina cyanea</i> ^c	39 (20)	33 (25)
Eastern Towhee, <i>Pipilo erythrophthalmus</i>	75 (38)	72 (55)
Field Sparrow, <i>Spizella pusilla</i> ^c	30 (15)	21 (16)
Chipping Sparrow, <i>S. passerina</i> ^c	20 (10)	20 (15)
Scarlet Tanager, <i>Piranga olivaceus</i>	30 (15)	28 (21)
Abundance, all species combined ^c	946 (480)	845 (644)
Total number of species	40	46

^a Population abundance significantly different between periods ($\chi^2 \geq 3.84$, $df = 1$, $P \leq 0.05$).

^b Population abundance slightly different between periods ($\chi^2 \geq 2.71$, $df = 1$, $0.10 \geq P \geq 0.05$).

^c Species observed only in the treated sector.

metella carolinensis), Chestnut-sided Warbler (*Dendroica pensylvanica*), and Hooded Warbler dramatically increased ($P \leq 0.05$) from 2001–2002 to 2005–2007 at the Barrens GHMA (Table 1). Populations of Tufted Titmice (*Baeolophus bicolor*) also had significant increases ($0.10 \geq P \geq 0.05$) from 2001–2002 to 2005–2007. No populations of early successional bird species declined ($P \geq 0.05$) over time despite increased forest maturation from 2001–2002 to 2005–2007. Populations of three species declined ($P \leq 0.05$) from 2001–2002 to 2005–2007: a woodland species, Ovenbird (*Seiurus aurocapilla*), a secondary forest species, American Redstart (*Setophaga ruticilla*), and a corvid, Blue Jay (*Cyanocitta cristata*).

DISCUSSION

I attributed a higher number of species in 2005–2007 on the Barrens GHMA, in part, to different levels of sampling between periods (e.g., 2 vs. 3 consecutive years, respectively). Perhaps the decline in abundance of all species combined was a result of the site being more heterogeneous in 2001–2002 than in 2005–2007.

Populations of Blue-winged Warbler may be increasing in central Pennsylvania at the expense of populations of Golden-winged Warbler (*Vermivora chrysoptera*) (Kubel 2005), but the Blue-winged Warbler was absent earlier on the area (Yahner 1997, 2003).

Population declines of Ovenbird at the Barrens GHMA in 2005–2007 paralleled significant Pennsylvania decreases in populations of this species according to the Breeding Bird Survey (BBS) (USDI 2007) during the period from 1980 to 2006. According to the BBS, American Redstart and Blue Jay populations in Pennsylvania have not declined significantly during this period. Blue Jays, however, may be experiencing localized population decreases due to West Nile virus (Male 2003, Rohnke 2005).

Gray Catbird populations increased in 2005–2007, but no similar trend was found in BBS data for Pennsylvania (USDI 2007). In contrast, populations of both Chestnut-sided and Hooded warblers increased significantly based on BBS data for Pennsylvania. I suspect that increases in populations on the area of these latter two species were due to regional

trends and were affected positively by presence of older habitat, especially in aspen D and in oak C plots on my study site. Chestnut-sided Warblers were especially abundant in these types of plots at the Barrens GHMA (Schill 2007).

My prediction (Yahner 2003) that early successional species would decline in 2005–2007 did not occur at the Barrens GHMA. It is probable that plant succession has not progressed adequately to affect population densities of these species, despite 5–7 years having elapsed since the last cutting cycle on the area. Populations of Field Sparrows (*Spizella pusilla*) were quite low at the Barrens GHMA prior to the fourth cutting cycle in 1999–2000; subsequent to this cycle, populations increased significantly (Yahner 2003) and have not changed since, based on results obtained in the present study. Habitat management for Ruffed Grouse on the Barrens GHMA generally did not have a profound effect on the sympatric bird community after the last cutting cycle. It is probable that more time must elapse before significant responses by bird populations will occur as a result of even-aged management at the Barrens GHMA.

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LITERATURE CITED

- GULLION, G. W. 1977. Forest manipulation for Ruffed Grouse. Transactions of the North American Wildlife and Natural Resources Conference 42:449–458.
- GULLION, G. W. 1990. Forest-wildlife interactions. Pages 349–383 in Introduction to forest science (R. A. Young and R. L. Giese, Editors). Second Edition. John Wiley and Sons, New York, USA.
- KUBEL, J. E. 2005. Breeding ecology of Golden-winged Warblers in managed habitats in central Pennsylvania. Thesis. Pennsylvania State University, University Park, USA.
- MALE, T. 2003. Potential impact of West Nile virus on American avifaunas. Conservation Biology 17: 928–930.
- REPENNING, R. W. AND R. F. LABISKY. 1985. Effects of even-age management on bird communities of the longleaf pine forest in northern Florida. Journal of Wildlife Management 49:1088–1098.
- ROHNKE, A. T. 2005. Possible ecological effects of habitat alteration and West Nile virus on songbird populations in central Pennsylvania. Thesis. Pennsylvania State University, University Park, USA.
- SCHILL, K. L. 2007. Habitat use and nesting ecology of Chestnut-sided Warblers. Thesis. Pennsylvania State University, University Park, USA.
- SOKAL, R. R. AND F. J. ROHLF. 1995. Biometry. Third Edition. W. H. Freeman and Company, New York, USA.
- STRELKE, W. K. AND J. G. DICKSON. 1980. Effect of forest clear-cut edge on breeding birds in east Texas. Journal of Wildlife Management 44:559–567.
- U. S. DEPARTMENT OF INTERIOR (USDI). 2007. Breeding Bird Surveys. USDI, Geological Survey, Patuxent Wildlife Research Center, Laurel, Maryland, USA. www.pwrc.usgs.gov/BBS/ (accessed 20 December 2007).
- YAHNER, R. H. 1987. Use of even-aged stands by winter and spring bird communities. Wilson Bulletin 99:218–232.
- YAHNER, R. H. 1993. Effects of long-term forest clear-cutting on wintering and breeding birds. Wilson Bulletin 105:239–255.
- YAHNER, R. H. 1997. Long-term dynamics of bird communities in a managed forested landscape. Wilson Bulletin 109:595–613.
- YAHNER, R. H. 2000. Long-term effects of even-aged management on bird communities in central Pennsylvania. Wildlife Society Bulletin 28:1102–1110.
- YAHNER, R. H. 2003. Responses of bird communities to early successional habitat in a managed landscape. Wilson Bulletin 115:292–298.